Mathematical Physics I (Fall 2026): Homework #3

Due Oct. 16, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts, turn in as a single pdf file to eTL]

• By turning in your homework, you acknowledge that you have not received any unpermitted aid, nor have compromised your academic integrity during its preparation. (Remember the SNU College of Natural Sciences Honor Code, and the SNU AI Guideline!)

• Textbook problem numbering convention (Boas, 3rd ed.): [Section].[Problem No.]. Many problems in this assignment are from the list of suggested problems announced in the class.

• Exhibit all intermediate steps to receive full credits. Only handwritten answers are accepted except for numerical problems or verifications — for which you print out and turn in not just the end results (e.g., plots) but also the source codes in your favorite tools such as MATLAB or MATHEMATICA.

1.-3. Boas Chapter 4, Problems 7.28, 8.8, 12.13

(Note: For Problem 7.28, consider ideal gas. You may want to prove and utilize the findings in Problem 7.27. Note the meaning of the subscripts next to the partial derivatives from Boas Chapter 4, Section 1 in case you did not read it. Prove further that the resulting formula is the familiar $c_p - c_v = nR$ found in e.g., Schroeder or Halliday & Resnick, where n is the number of moles of gas present and R is the gas constant. For Problem 8.8, tackle the stated problem in two ways, one with the method in Section 8 and the other with the Lagrange multiplier described in Section 9.)

4.-5. Boas Chapter 5, Problems 3.7, 5.1

(Note: For Problem 3.7, review the notations introduced in the preceding pages. For (d), review and utilize the two theorems discussed in the current section.)

6.-8. Boas Chapter 6, Problems 3.12, 6.16, 9.4

9. A localized, axisymmetric charge density distribution is given by $\rho(r,\theta) = \rho_0 \frac{\cos\theta}{r}$ within the region $a \le r \le b$ centered on the origin, where θ is measured from the z-axis.

(a) Calculate the net enclosed charge $q = \int \rho(r,\theta) dV$ over the region.

(b) Compute the total z-component of the dipole moment $p_z = \int (r\cos\theta)\rho(r,\theta) dV$.

Now suppose we are interested in the electric field far from a localized charge distribution. In this limit, the axisymmetric electrostatic potential is written as:

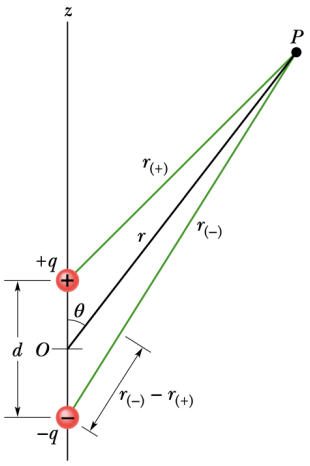
$$V(r,\theta) = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r} + \frac{p\cos\theta}{r^2}\right)$$

where q is the monopole charge and p is the dipole moment.

(c) Compute the electric field components in polar coordinates, (E_r, E_θ).

(d) For a pure dipole ($q = 0$), find all angles $\theta \in [0, \pi]$ where the radial component E_r vanishes.

(e) Using the multivariable chain rule, evaluate $\frac{dV}{dt}$ along the path $(r(t), \theta(t)) = (r_0 e^{\alpha t}, \omega t)$.



[Adapted from Halliday & Resnick, Chapter 24.4]

(Note: To learn more about the electric dipole, you may want to review the freshman physics textbooks such as Halliday & Resnick, or the textbooks in electromagnetism such as Griffiths.)

10. (a) Review Boas Chapter 6, Example 7.2, thus prove Eq.(7.6) or Eq.(f) in p.339. You are also asked to prove the identity by explicitly working with the components, i.e.,

$$\nabla \cdot (\phi \mathbf{V}) = \frac{\partial(\phi V_x)}{\partial x} + \frac{\partial(\phi V_y)}{\partial y} + \frac{\partial(\phi V_z)}{\partial z} = \ldots$$

(b) Review how Gauss's law $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ is acquired from Coulomb's law in Boas Chapter 6, Section 10, where $\mathbf{E}$ is the electric field, ρ is the volume charge density, and ϵ_0 is the permittivity of free space.

(c) Using (a) and (b), show that the energy of a continuous charge distribution is given by

$$W_\mathrm{e} = \frac{1}{2}\int \rho V d\tau = \frac{\epsilon_0}{2}\int |\mathbf{E}|^2 d\tau$$

for integration over all space, where V is the electric potential. You are first asked to briefly discuss how one arrived at the term $W_\mathrm{e} = \frac{1}{2}\int \rho V d\tau$, and then explicitly prove the second equality. You may assume that V vanishes at large distance r at least as fast as r^{-1}.

(d) Now, prove Eq.(h) in p.339. You may prove the identity by explicitly working with the components, i.e.,

$$\nabla \cdot (\mathbf{U} \times \mathbf{V}) = \frac{\partial (\mathbf{U} \times \mathbf{V})_x}{\partial x} + \frac{\partial (\mathbf{U} \times \mathbf{V})_y}{\partial y} + \frac{\partial (\mathbf{U} \times \mathbf{V})_z}{\partial z} = ...$$

(e) Review how Ampere's law $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ is acquired from the Ampere's circuital law in Boas Chapter 6, Section 11, where $\mathbf{B}$ is the magnetic field, $\mathbf{J}$ is the current density, and μ_0 is the permeability of free space.

(f) Using (d) and (e), show that the energy stored in magnetic fields is given by

$$W_{\mathrm{m}} = \frac{1}{2} \int (\mathbf{A} \cdot \mathbf{J}) d\tau = \frac{1}{2\mu_0} \int |\mathbf{B}|^2 d\tau$$

for integration over all space, where $\mathbf{A}$ is the magnetic vector potential. You are first asked to briefly discuss how one arrived at the term $W_{\mathrm{m}} = \frac{1}{2} \int (\mathbf{A} \cdot \mathbf{J}) d\tau$, and then explicitly prove the second equality. You may assume that $\mathbf{A}$ vanishes at large distance r at least as fast as r^{-1}.

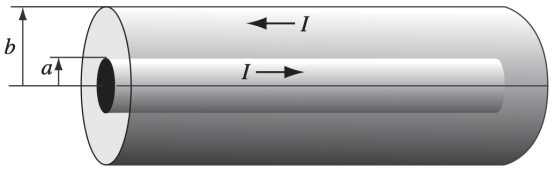
(g) Finally, using Maxwell's equations in vacuum with sources,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$

and the identity found in (d), derive Poynting's theorem:

$$\mathbf{E} \cdot \mathbf{J} = -\frac{\partial u}{\partial t} - \nabla \cdot \mathbf{S},$$

where $u = \frac{1}{2} \left(\epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right)$ is the electromagnetic energy density and $\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B})$ is the Poynting vector. Briefly state the physical interpretation of the theorem.



[Adapted from Griffiths, Chapter 7.2]

(Note: The exercises here should sound familiar to most of you as you have begun to study and explore electromagnetism. If not, you may want to briefly review the classic textbooks in electromagnetism such as Griffiths (Chapters 2, 7, and 8) or Jackson. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see http://library.snu.ac.kr/using/proxy.)